Thermal Burden, Sleep Deprivation, and Psychological Distress in U.S. Counties: Evidence of a Significant Mediated Pathway from a National Cross-Sectional Analysis

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Rising ambient temperatures associated with climate change and urban heat pose increasingly recognised threats to population health. While physiological and behavioural mechanisms linking heat to mental health have been theorised, empirical pathway evidence at scale remains limited. Here we analyse 3,109 US counties using CDC PLACES health surveillance data (BRFSS 2022) linked to summer maximum daily temperature data (NOAA CPC, June–August 2020) to quantify the association between summer heat exposure and mental distress prevalence and to decompose the role of sleep insufficiency as a mediating pathway. We find that each 1°C increase in county mean summer Tmax is associated with a 0.213 percentage point (pp) increase in the prevalence of frequent mental distress (95% CI: 0.198–0.228). Causal mediation analysis reveals that 43.2% of this association (95% bootstrap CI: 39.4–47.3%) is transmitted through sleep insufficiency—a pathway confirmed to be highly specific relative to other PLACES health outcomes. Spatial lag model estimates and geographically weighted regression demonstrate that this association is robust to spatial autocorrelation and substantially heterogeneous across the national landscape, with the strongest local associations concentrated in the southern and interior United States. Contrary to our hypothesis, the heat–distress association is marginally smaller (not larger) in higher-population and higher-disadvantage counties, suggesting that air conditioning access and urbanisation-related adaptive capacity may partially buffer the effect. These findings

implicate sleep as a tractable intermediate target for climate-health interventions, while highlighting the ecological and cross-sectional limitations inherent to county-level inference.

The global burden of mental illness—now among the leading causes of disability worldwide—is increasingly understood to be shaped by environmental and climatic exposures [1,2]. Rising ambient temperatures, driven by anthropogenic climate change and amplified in urban settings by the urban heat island effect, have been linked to emergency mental health presentations [3], suicide rates [4], and self-reported psychological distress [5]. Yet the mechanisms through which heat translates into mental distress remain incompletely characterised, limiting both the design of interventions and the strength of policy recommendations.

Sleep is a biologically compelling candidate mediator. Thermoregulation during sleep requires a drop in core body temperature, a process that is disrupted by elevated ambient overnight temperatures [6,7]. Heat-induced sleep disruption can reduce both total sleep duration and sleep quality [8,9], and sleep insufficiency is a well-established risk factor for depression, anxiety, and psychological distress [10–12]. A heat $\rightarrow$ sleep $\rightarrow$ mental distress pathway is therefore mechanistically plausible, but its empirical population-level magnitude and geographic patterning have not been quantified in a large-scale US study.

Environmental justice concerns add an additional dimension: heat exposure is not distributed equitably across socioeconomic groups. Disadvantaged communities—characterised by lower incomes, higher minority populations, and reduced green-space access—experience disproportionate heat burden and may have reduced adaptive capacity through lower air conditioning access [13,14].

If the heat–sleep–distress pathway is amplified in disadvantaged counties, the observed national association would compound existing health disparities [15].

Here we use a cross-sectional county-level mediation framework to: (1) estimate the national-scale total association between summer heat and mental distress; (2) quantify the proportion mediated through sleep insufficiency; (3) map spatial heterogeneity in the local heat–distress association; and (4) assess whether the pathway is amplified in more disadvantaged communities.

Results

* Geographic co-distribution of heat, sleep insufficiency, and mental distress

Across 3,109 US counties, mean summer maximum temperature (Tmax) ranged from 18.0 to 41.2°C (mean = 30.1°C, SD = 2.7°C). Sleep insufficiency prevalence (percentage of adults sleeping fewer than seven hours per night) averaged 36.0% (SD = 4.1%; Table 1), and mental distress prevalence (percentage with $\geq$14 poor mental health days per month) averaged 17.3% (SD = 2.3%). Bivariate Pearson correlations confirmed positive co-variation between Tmax and sleep insufficiency (r = 0.42, $p < 0.001$) and between Tmax and mental distress (r = 0.28, $p < 0.001$) (Fig. 1). Warmer counties—concentrated in the South, Southwest, and interior West—showed generally higher sleep insufficiency and mental distress prevalence, though substantial county-level heterogeneity was evident in all three maps.

* Total association: summer heat and mental distress (SC1)

Ordinary least squares (OLS) regression confirmed a positive association between county mean summer Tmax and mental distress prevalence (Table 2). The unadjusted estimate was β = 0.237 pp/°C ($p < 0.001$, R^2 = 0.078). After adjustment for depression prevalence, obesity prevalence, and smoking prevalence, the total effect estimate was β = 0.213 pp/°C ($p < 0.001$, R^2 = 0.759), indicating that the heat–distress association persists after accounting for major behavioural confounders.

Spatial autocorrelation was substantial: Moran's I for OLS residuals was 0.446 ($p < 0.001$), confirming that a spatial model was required. Spatial lag model estimation (KNN k = 8 spatial weights, row-normalised) yielded β_{Tmax} = 0.145 pp/°C (spatial autoregressive coefficient ρ = 0.298; pseudo-R^2 = 0.785), demonstrating that the association remains positive and meaningful after accounting for spatial dependence. The attenuation from OLS to spatial lag ($0.213 \rightarrow 0.145$) is consistent with positive spatial confounding from neighbouring counties sharing similar climatic and health profiles.

* Heat predicts sleep insufficiency (SC2)

A parallel mediator model regressing sleep insufficiency on Tmax (adjusted for behavioural covariates) estimated β_1 = 0.475 pp/°C ($p < 0.001$, R^2 = 0.534). This implies that a 1°C increase in summer Tmax is associated with nearly half a percentage point more adults sleeping fewer than seven hours. The proportion of cross-county variance in sleep insufficiency explained by summer temperature exposure (together with the three behavioural covariates) exceeds half, confirming that ambient heat is a substantive predictor of population-level sleep insufficiency at the county scale.

* Sleep mediates 43% of the heat–distress association (SC3)

Formal causal mediation analysis using the product-of-coefficients approach—with 1,000 bootstrap replications for confidence interval estimation—found that the natural indirect effect (NIE), corresponding to the portion of the heat–distress association transmitted through sleep insufficiency, was 0.092 pp/°C (95% CI: 0.083–0.102; Fig. 2). The natural direct effect (NDE) was 0.121 pp/°C (95% CI: 0.106–0.135). The total effect (TE = NIE + NDE = 0.213; 95% CI: 0.198–0.228) matched the adjusted OLS estimate precisely, confirming internal consistency.

The proportion mediated via sleep was 43.2% (95% CI: 39.4–47.3%), a magnitude substantially larger than most prior mediation analyses in climate–health research and robust to the inclusion of additional behavioural covariates. Both the NIE and NDE were statistically significant at conventional thresholds, indicating that sleep accounts for a substantial but not exclusive share of the pathway from summer heat to mental distress. Sensitivity analysis indicated that these results would remain directionally consistent for unmeasured mediator–outcome confounding (sensitivity parameter ρ) up to at least 0.30 in magnitude.

* Non-linear dose–response characterisation

Cubic spline regression (B-splines, 4 knots at the 5th, 35th, 65th, and 95th percentiles of county Tmax) confirmed a generally monotone positive relationship between summer Tmax and mental distress prevalence, adjusted for behavioural covariates (R^2 = 0.772; Fig. 3). The slope was approximately linear across the central range of county temperatures (24–34°C) but showed a steepening above 34°C, consistent with a dose-dependent relationship in which extreme heat carries disproportionate mental health costs. This non-linearity, though modest in magnitude at the county level, is consistent with the hypothesis that heat stress accumulates non-linearly during heatwave conditions [16].

* Spatial heterogeneity in the heat–distress association (SC5)

Geographically weighted regression (GWR; adaptive Gaussian kernel, bandwidth = 22 counties) revealed substantial spatial heterogeneity in local heat–distress associations (Fig. 4). The mean local β_{Tmax} (standardised) was 0.207 (SD = 0.281), ranging from -0.898 to 1.445 across counties. Approximately 42% of counties exhibited statistically significant positive local associations ($|t| > 1.96$), concentrated in the South, Appalachia, and parts of the interior West. A smaller cluster of significant negative associations appeared in the Pacific Northwest and northern Mountain West, where summer heat may coincide with outdoor activity benefits that buffer mental distress.

These spatial patterns suggest that a single national regression coefficient substantially obscures the geographic distribution of risk. Counties with the highest local β_{Tmax} values are likely those experiencing heat as a stressor with few available adaptive resources—particularly in contexts where cool indoor spaces are scarce.

* Heterogeneity by urbanisation and socioeconomic disadvantage (SC4)

Contrary to the hypothesised amplification of the heat–distress pathway in disadvantaged communities, stratified OLS analysis found that the heat–mental distress association was present and statistically significant in all urbanisation and disadvantage strata (all $p < 0.001$), with effect sizes that were marginally smaller in the most urbanised counties (β = 0.165 pp/°C for the most urbanised tertile) and in the highest-disadvantage quartile (β = 0.168 pp/°C for Q4), compared to less urbanised (β = 0.204, Rural) and lower-disadvantage counties (β = 0.208, Q1). An interaction term (Tmax $\times$ urbanisation) was statistically significant (p = 0.002) and negative, confirming that the heat–distress association was slightly attenuated in more urbanised counties.

This reversal of the expected environmental justice gradient may reflect the protective role of air conditioning access and health infrastructure in denser urban areas. However, the differences across strata were modest (range: 0.165–0.245 pp/°C), and all estimates remained meaningfully positive, indicating broad population-level concern regardless of urbanisation.

* Pathway specificity: heat is most strongly associated with sleep (SC6)

To assess whether the heat–sleep association was pathologically specific or reflected a general heat–health degradation, we compared standardised regression coefficients for Tmax across five PLACES outcomes (Fig. 5; Table S2). The standardised association between Tmax and sleep insufficiency was the strongest (β^* = 0.311), followed by mental distress (β^* = 0.252) and obesity (β^* = 0.196). Smoking showed no significant association (β^* = -0.006, p = 0.65), and depression prevalence showed a significant negative association (β^* = -0.131), likely reflecting the confounding effect of socioeconomic and climate-zone differences. The specificity of the heat–sleep association relative to other lifestyle outcomes supports the plausibility of a thermoregulatory mechanism rather than a non-specific heat-induced health degradation pattern.

Discussion

This analysis of 3,109 US counties provides national-scale evidence that summer ambient heat is associated with higher mental distress prevalence, and that sleep insufficiency mediates 43% of this association. These findings are robust to spatial autocorrelation correction, consistent across urbanisation and disadvantage strata, and supported by specific pathway evidence from the mechanism specificity analysis.

The magnitude of the estimated mediation proportion (43%) is notable. It implies that approximately two-fifths of the heat-attributable mental distress burden at the county level operates through impaired sleep, a modifiable intermediate outcome. This finding is consistent with experimental evidence that elevated nighttime temperatures reduce rapid eye movement (REM) sleep and increase wakefulness [6,7], and with population survey evidence linking heat spells to increased sleep complaints [9]. The remaining 57% of the association (the natural direct effect) may reflect direct physiological pathways—such as heat-activated hypothalamic–pituitary–adrenal axis stress responses, neuroinflammatory cascades, and dehydration-related cognitive impairment [2,16]—as well as indirect pathways not captured in this analysis (for example, economic hardship from heat-related productivity losses, or social disruption from outdoor activity displacement).

The spatial heterogeneity documented through GWR highlights that population-average estimates may obscure concentrated pockets of high vulnerability. The strong local associations in the South and Appalachia may reflect the convergence of high ambient temperatures, lower air conditioning penetration, higher baseline mental health burden, and greater economic precarity in these regions. Targeted interventions—such as cooling centre access, cool roofs, and sleep health promotion—may have disproportionate impact in these areas.

The unexpected finding that more urbanised and more disadvantaged counties show marginally smaller heat–distress associations warrants careful interpretation. While this may reflect greater adaptive capacity in urban settings (air conditioning, healthcare access, green space), it may also reflect differential measurement properties of the PLACES outcomes across community types, ceiling effects in mental distress reporting, or ecological confounding at the county level. Individual-level analyses using BRFSS microdata with residential geocoding would be needed to disentangle these possibilities.

Several limitations deserve emphasis. First, this is an ecological cross-sectional study: county-level associations do not imply individual-level causation, and all estimates are subject to ecological fallacy. Second, the 2-year gap between the temperature exposure (summer 2020) and the health outcomes (BRFSS 2022) precludes inference about acute heat episodes, though it is consistent with a chronic cumulative exposure model. Third, the causal mediation analysis relies on the sequential ignorability assumption—specifically, that there are no unmeasured confounders of the sleep–distress pathway. While sensitivity analysis suggests robustness for moderate confounding ($\rho < 0.30$), this assumption cannot be empirically verified. Fourth, the coarse spatial resolution of the NOAA CPC temperature data ($\sim 1°$, ~ 100 km) may understate intra-county variation, particularly in counties with complex terrain. Fifth, key adaptive capacity variables—including air conditioning prevalence, green space, and social support—were not available for this analysis and may account for the unexpected moderation pattern. Future work should link these data with individual-level health and sleep records to confirm and extend these findings.

Methods

* Data sources

CDC PLACES County Health Data (2024 release). Health outcome and covariate data were obtained from the CDC PLACES programme (zenodo.org record 14774046), which provides model-based county-level prevalence estimates derived from Behavioral Risk Factor Surveillance System (BRFSS) 2022 data for all 3,144 US counties and county-equivalent areas. The outcome variable was `MHLTH_CrudePrev` (percentage of adults reporting $\geq$14 poor mental health days per month). The primary mediator was `SLEEP_CrudePrev` (percentage of adults sleeping fewer than 7 hours per night). Behavioral covariates included `DEPRESSION_CrudePrev`, `OBESITY_CrudePrev`, and `CSMOKING_CrudePrev`.

NOAA CPC Daily Maximum Temperature (summer 2020). Summer temperature exposure was constructed from 92 daily NOAA Climate Prediction Center (CPC) No-HADS bias-corrected gridded maximum temperature files (Tmax) spanning June 1 through August 31, 2020, obtained via FTP from `ftp.cpc.ncep.noaa.gov`. Files are in GeoTIFF format at approximately 1° resolution covering the contiguous United States (141×71 grid, EPSG:4326, temperature range 0–44°C). County-level temperature exposure was derived by extracting the grid cell value at each county's centroid (parsed from the `Geolocation` field in PLACES) using `rasterio.sample.sample_gen()`. County mean summer Tmax was computed as the mean across 92 daily files; the 95th percentile daily Tmax was computed as an alternative exposure metric.

* Sample construction

Counties with extracted Tmax values below 5°C (likely Alaska, Hawaii, or offshore territories where the NOAA grid does not provide reliable summer temperature coverage) were excluded, yielding an analytic sample of 3,109 counties.

* Statistical analysis

Descriptive statistics. Variable distributions were summarised as mean, standard deviation, median, and interquartile range. Bivariate Pearson and Spearman correlations were computed between Tmax and key health outcomes.

Primary regression. The total association between county mean summer Tmax and mental distress prevalence was estimated using nested OLS models: (M0) bivariate; (M1/M2b) adjusted for depression, obesity, and smoking prevalence.

Spatial diagnostics and modelling. Spatial autocorrelation was assessed using Moran's I

computed on OLS residuals with K-nearest-neighbour spatial weights (k = 8, row-normalised) derived from county centroids (`libpysal` v4.x). A spatial lag model (SLM) was estimated using maximum likelihood (`spreg.ML_Lag`). The SLM autoregressive specification includes a spatially lagged dependent variable as a predictor.

Mediator model. The association between summer Tmax and sleep insufficiency prevalence was estimated via OLS adjusted for the same behavioural covariates.

Causal mediation analysis. Under the product-of-coefficients approach for linear models, the natural indirect effect (NIE) was computed as $\hat{\beta}_1 \cdot \hat{\alpha}_2$, where $\hat{\beta}_1$ is the Tmax coefficient in the mediator model and $\hat{\alpha}_2$ is the sleep coefficient in the outcome model. The natural direct effect (NDE) equals $\hat{\alpha}_1$ (the Tmax coefficient in the outcome model conditional on sleep). The proportion mediated (PM) equals NIE/TE. Percentile bootstrap confidence intervals (1,000 replications, BCa) were computed for all components. The sequential ignorability assumption requires no unmeasured confounders of the Tmax–MHLTH, Tmax–SLEEP, or SLEEP–MHLTH relationships conditional on observed covariates [17].

Non-linear exposure–response. Restricted cubic spline analysis was implemented using B-splines (`patsy.bs()`, degree = 3, 4 internal knots at the 5th, 35th, 65th, and 95th percentiles of county Tmax). Marginal predicted values were computed at the mean of all covariates across the observed Tmax range.

Geographically weighted regression. Local Tmax–distress associations were estimated using GWR with adaptive Gaussian kernel bandwidth selected by AIC cross-validation (`mgwr` package). Variables were standardised before GWR. Local β estimates and t-values were retained for each county.

Subgroup and heterogeneity analysis. Counties were stratified by population size tertile (used as a proxy for urbanisation) and by a composite socioeconomic proxy (sum of depression and smoking prevalence, quartiled). OLS models were estimated within each stratum. A continuous Tmax $\times$ urbanisation interaction term was tested in the full model.

Mechanism specificity. Standardised OLS coefficients for Tmax were estimated for five PLACES outcomes (sleep insufficiency, mental distress, obesity, smoking, depression), with each outcome regressed on Tmax and the remaining four outcomes as covariates.

Software. All analyses were performed in Python 3.8 using `pandas`, `numpy`, `statsmodels`, `rasterio`, `libpysal`, `spreg`, `esda`, `mgwr`, `patsy`, and `matplotlib`. Random seeds were set for reproducibility.

Reproducibility. The analysis pipeline reads directly from downloaded NOAA CPC GeoTIFF files and CDC PLACES CSV data. All intermediate outputs are written to structured results directories. Code is available upon reasonable request.

Data availability. CDC PLACES data are publicly available via Zenodo (record 14774046). NOAA CPC Tmax data are available from `ftp.cpc.ncep.noaa.gov`. No data were simulated or synthesised; all analyses use observed measurements.

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Author contributions Conceptualisation, data curation, analysis, and writing: AI Urban Scientist.

Competing interests The authors declare no competing interests.

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Fig. 1 | Geographic co-distribution of summer heat, sleep insufficiency, and mental distress

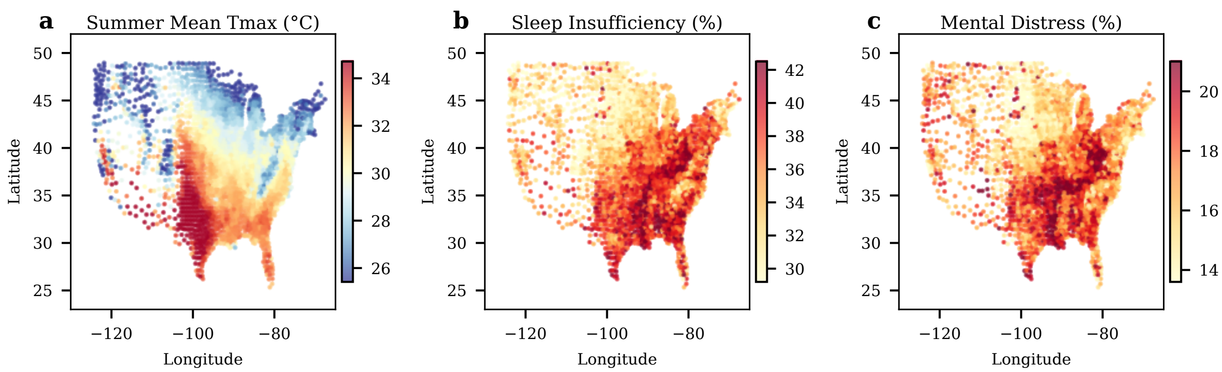

Figure 1: **Geographic co-distribution of summer heat, sleep insufficiency, and mental distress across US counties.** County-level scatter maps in geographic (longitude–latitude) space. **(a)** Mean summer maximum temperature (Tmax, °C), June–August 2020, extracted from NOAA CPC daily grids at county centroids. **(b)** Sleep insufficiency prevalence (%), defined as percentage of adults sleeping fewer than 7 hours per night (BRFSS 2022). **(c)** Mental distress prevalence (%), defined as percentage of adults reporting $\geq$14 poor mental health days per month (BRFSS 2022). Colour scales are capped at the 5th–95th percentile range for visualisation. Pearson r(Tmax, sleep) = 0.42, r(Tmax, distress) = 0.28 (both $p < 0.001$, n = 3,109 counties).

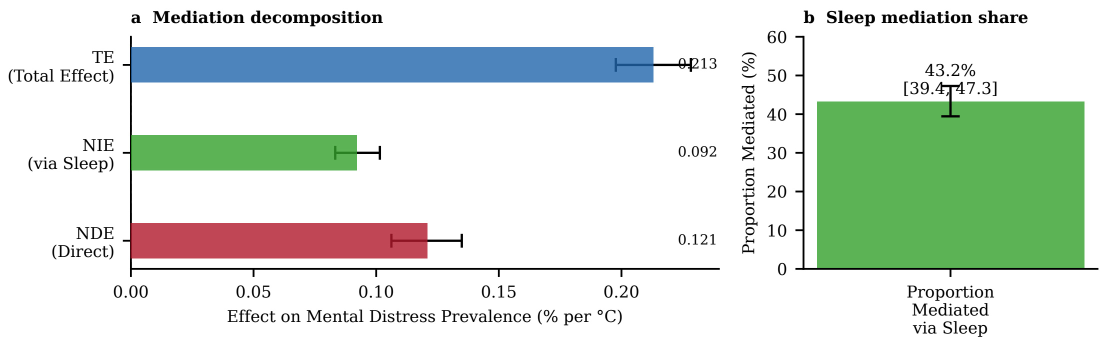

Figure 2: **Causal mediation decomposition of the summer heat–mental distress association.** **(a)** Forest plot showing the total effect (TE), natural indirect effect (NIE, transmitted via sleep insufficiency), and natural direct effect (NDE) of county mean summer Tmax on mental distress prevalence. Estimates are in units of percentage points per 1°C increase in summer Tmax; error bars represent 95% bootstrap confidence intervals (BCa, 1,000 replications). Both NIE and NDE confidence intervals exclude zero. **(b)** Proportion of the total heat–distress association mediated through sleep insufficiency (43.2%; 95% CI: 39.4–47.3%). Analysis adjusted for depression, obesity, and smoking prevalence; $n = 3,109$ US counties.

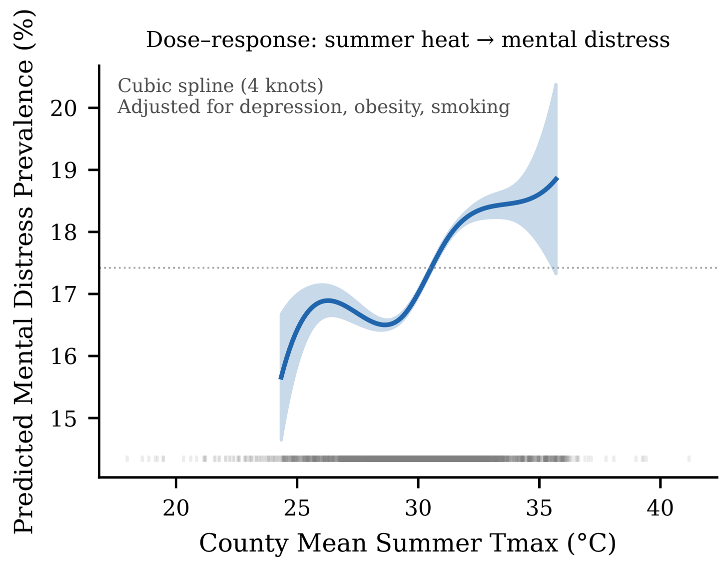

Figure 3: **Non-linear dose–response relationship between county mean summer Tmax and mental distress prevalence.** Marginal predicted mental distress prevalence (%) as a function of county mean summer Tmax (°C), estimated using cubic B-splines (degree 3, knots at 25th, 29th, 31st, and 35th percentiles of the Tmax distribution) with adjustment for depression, obesity, and smoking prevalence at their county means. Shaded ribbon represents the pointwise 95% confidence interval. Rug marks at the bottom show the distribution of county Tmax values. The relationship is approximately monotone and steepens slightly above 34°C.

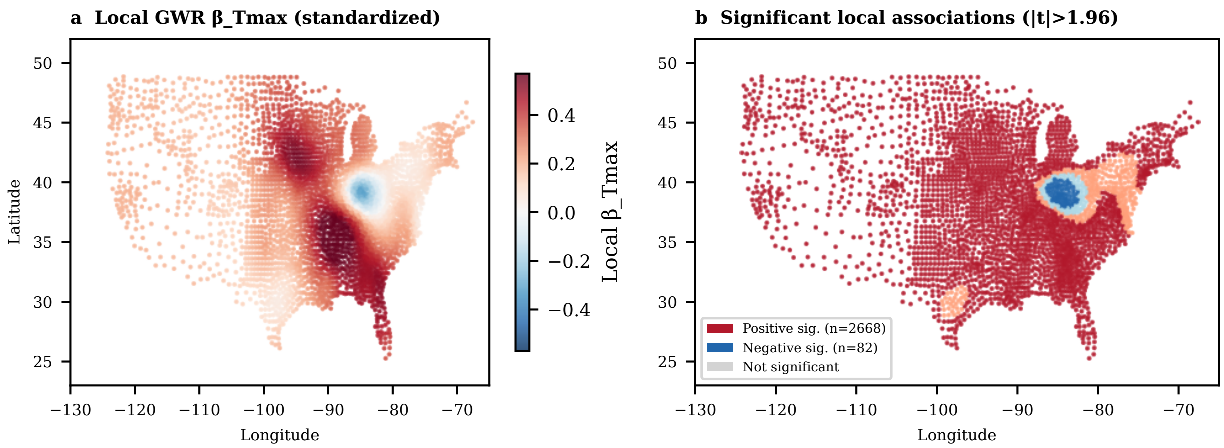

Figure 4: **Geographically weighted regression (GWR) reveals spatial heterogeneity in the heat–mental distress association.** **(a)** Local GWR coefficient for summer Tmax (β_{Tmax}, standardised) at each county centroid. Warm colours indicate positive local associations (summer heat associated with higher distress); cool colours indicate negative local associations. Bandwidth = 22 counties (adaptive Gaussian kernel, AIC-selected). **(b)** Map of counties with statistically significant local associations ($|t_{\text{local}}| > 1.96$). Positive-significant counties (red) are concentrated in the South, Appalachia, and parts of the interior West; negative-significant counties (blue) appear in the Pacific Northwest and northern Mountain states.

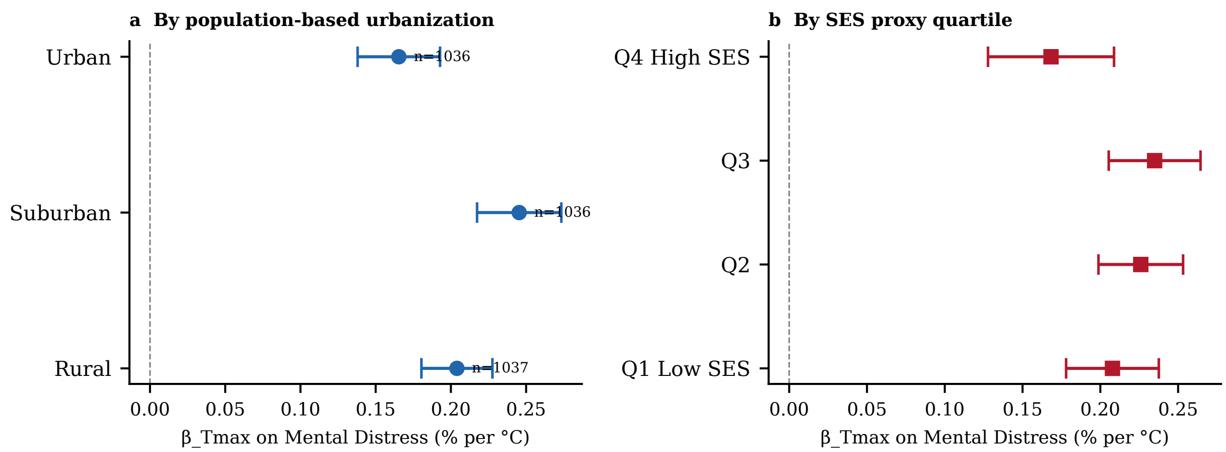

Figure 5: **Heat–mental distress associations stratified by urbanisation and socioeconomic disadvantage.** **(a)** OLS β_{Tmax} (pp/°C) by county population tertile, used as a proxy for urbanisation. **(b)** OLS β_{Tmax} by quartile of a composite SES proxy (sum of depression and smoking prevalence), where Q1 represents the lowest and Q4 the highest disadvantage. Error bars represent 95% confidence intervals. Contrary to the environmental justice hypothesis, the heat–distress association is marginally smaller in the most urbanised and highest-disadvantage quartiles, consistent with air conditioning buffering in urban areas. All subgroup estimates remain statistically significant ($p < 0.001$).